MIL-I-23053/4C
01 May 1986
SUPERSEDING
MIL-I-23053/4B
20 May 1976

MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, POLYOLEFIN, DUAL-WALL, OUTER WALL CROSSLINKED

This specification has been approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-I-23053.

REQUIREMENTS:

Continuous operating temperature range: -55°C (-67°F) to +110°C (+230°F).

Composition: The heat shrinkable sleeving shall consist of a crosslinked outer wall which shrinks and an inner wall which flows upon the application of heat.

Color: The heat shrinkable sleeving shall be furnished in the color specified by the acquisition activity (see 6.2.1); however, black shall be considered standard. Colors shall conform to the requirements of Class 1 of MIL-STD-104.

Classification: The heat shrinkable sleeving shall be furnished in the following classes, as specified (see 6.2.1):

Class 1 - Semi-rigid, non-flame retardant
Class 2 - Flexible, flame retardant
Class 3 - Flexible, flame retardant, high expansion

Longitudinal change:

Class 1 - +1, -10 percent
Class 2 - +1, -5 percent
Class 3 - +1, -15 percent

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:

NOTE: Entire specification sheet revised.

AMSC N/A FSC 5970

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TABLE I. Construction details, inches (mm). 1/

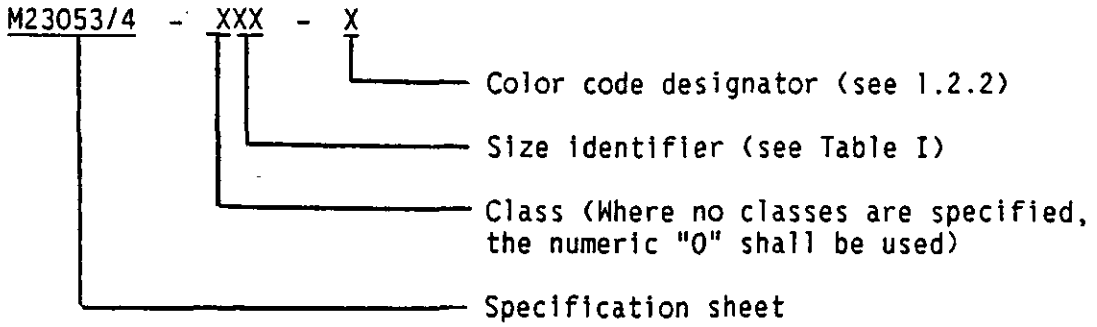
Military part number 3/	As supplied ID minimum	After unrestricted shrinkage		Inner meltable wall thickness, nominal
		ID maximum	Total wall thickness 1/	
Class 1				
M23053/4-101-*	.125 (3.2)	.023 (.58)	.038 ± .006 (.97 ± .152)	.020 (.508)
M23053/4-102-*	.187 (4.7)	.060 (1.52)	.043 ± .006 (1.09 ± .152)	.025 (.635)
M23053/4-103-*	.250 (6.4)	.080 (2.03)	.047 ± .007 (1.19 ± .178)	.027 (.686)
M23053/4-104-*	.375 (9.5)	.135 (3.43)	.050 ± .007 (1.27 ± .178)	.030 (.762)
M23053/4-105-*	.500 (12.7)	.195 (4.95)	.055 ± .007 (1.40 ± .178)	.035 (.889)
M23053/4-106-*	.750 (19.1)	.313 (7.95)	.065 ± .007 (1.65 ± .178)	.040 (1.016)
M23053/4-107-*	1.000 (25.4)	.400 (10.2)	.075 ± .007 (1.91 ± .178)	.040 (1.016)
M23053/4-108-*	.300 (7.6)	.050 (1.27)	.100 ± .008 (2.54 ± .203)	.065 (1.651)
Class 2				
M23053/4-201-*	.238 (6.06)	.125 (3.18)	.029 ± .005 (7.37 ± .127)	.004 (.102)
M23053/4-202-*	.355 (9.02)	.187 (4.75)	.029 ± .005 (7.37 ± .127)	.004 (.102)
M23053/4-203-*	.475 (12.07	.250 (6.35)	.030 ± .005 (7.62 ± .127)	.005 (.127)
M23053/4-204-*	.712 (18.08)	.375 (9.53)	.035 ± .005 (8.89 ± .127)	.005 (.127)
M23053/4-205-*	.950 (24.13)	.500 (12.7)	.042 ± .007 (10.7 ± .178)	.007 (.178)
M23053/4-206-*	1.425 (35.20)	.750 (19.05)	.047 ± .008 (11.9 ± .203)	.007 (.178)
Class 3				
M23053/4-301-*	.120 (3.05)	.040 (1.02)	.040 ± .010 (1.02 ± .254)	.020 (.508)
M23053/4-302-*	.240 (6.10)	.080 (2.04)	.040 ± .010 (1.02 ± .254)	.020 (.508)
M23053/4-303-*	.470 (11.94)	.160 (4.06)	.070 ± .014 (1.78 ± .356)	.030 (.762)
M23053/4-304-*	.940 (23.87)	.320 (8.13)	.100 ± .020 (2.54 ± .508)	.040 (1.02)
M23053/5-305-*	1.570 (39.38)	.510 (12.95)	.100 ± .020 (2.54 ± .508)	.040 (1.02)

1/ Diameter limits for objects to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

3/ The asterisk in the part number shall be replaced by color code designations (see 1.2.2).

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Example: Class 1, black sleeving .750 inch (19.1 mm) as supplied ID sleeving shall be identified as M23053/4-106-0.

Unrestricted shrinkage: Test method 4.5.5.2; 200° ± 2°C (392° ± 4°F) for 10 minutes.

TABLE II. Physical properties.

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Cold impact	No cracking	4.6.7.2 ASTM D 746, -55° ± 2°C (-67 ± 4°F)
Heat shock	No cracks, flowing or dripping of outerwall	4.6.8 250° ± 3°C (482° ± 6°F) 1/
Sealing efficiency (Classes 1 and 3 only)	No openings on reheat	2/
Secant modulus, psi (MPa)	Class 1 - 25,000 (172) minimum Class 2 - 25,000 (172) maximum Class 3 - 12,000 (83) maximum	4.6.13.1, ASTM D 882, 2 percent strain
Color stability	Pass	4.6.16 175° ± 2°C (347° ± 4°F), 24 hours

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength, psi (MPa), minimum	1,500 (103) - Classes 1 & 2 1,300 (9.0) - Class 3	4.6.14, ASTM D 638, 2 inches/minute
Ultimate elongation, percent, minimum	200 - Classes 1 & 2 250 - Class 3	4.6.14, ASTM D 638, 2 inches/minute
Dielectric strength, volts/mil (Kv/mm), minimum	500 (19.7) - Classes 1 & 2; 300 (11.8) - Class 3	4.6.2, ASTM D 2671
Volume resistivity, Ohm-CM, minimum	1×10^{15} -Class 1; 1×10^{12} - Classes 2 & 3	4.6.2 ASTM D 876
Flammability	Class 1 - N/A; Classes 2 & 3 - self-extinguishing within one minute, no more than 25 percent of indicator flag burned or charred	4.6.15.1
Corrosion	No corrosion	4.6.10.1 and 4.5.10.2 121 ± 2°C (250 ± 4°F) for 16 hours
Water absorption, percent, maximum	0.1 - Class 1 0.5 - Class 2 1.0 - Class 3	4.6.2 ASTM D 570, 24 hrs at 23°C
Heat resistance, properties after:		4.6.9 3/
Visual inspect and bend test	No dripping, flowing of outer wall. No cracks	

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TABLE II. Physical properties. - Continued

Characteristic	Requirement	Test procedure and conditions
Fluid resistance, properties after:		4.6.11
Tensile stength, psi (MPa), minimum	1,000 (6.9) - Classes 1 & 2 900 (6.2) - Class 3	
Dielectric strength, volts/mil (Kv/mm), minimum	400 (15.8) - Classes 1 & 2 200 (7.9) - Class 3	
Fungus resistance 4/	No growth	4.6.2, ASTM G 21

1/ Use mandrel sizes specified in footnote 3/

2/ Sealing Efficiency: A six-inch length of sleeving shall be heated in an oven at 200° ± 2°C (392° ± 4°F) for 3 minutes. Within 5 seconds after removal, approximately .25 inch (6.35 mm) of one end shall be lightly pressed together with a pair of pliers and pressure held for 20 to 40 seconds. Allow the sleeving to condition at room temperature for 10 minutes. Reheat the sleeving in the 200°C oven for 5 minutes. Remove from the oven and examine for openings through the pressed or sealed area of the sleeving.

3/ 175° ± 2°C (347° ± 4°F) for 168 hours followed by a mandrel bend test at room temperature. The sleeving shall be bent through 360°in 10 ± 2 seconds. Mandrel size shall be as follows:

Nominal sleeving ID, inch (mm) minimum, as supplied	Mandrel diameter, inch (mm)
.125 to .250 (3.18 to 6.35)	.438 (11.1)
over .250 to .750 (6.35 to 19.1)	.50 (12.7)
over .750 (19.1)	.563 (14.3)

4/ MIL-I-23053/4 materials do not normally support fungus growth. Performance of this requirement is only when specified by the acquisition activity (see 6.2.1).

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Shelf life conditions: Supplier shall certify to storage at 40°C (104°F) for 2 years. Conformance to 3.5. See 3.5.2 for shelf life extension.

NOTICE

a. Flammability - Class 1 shrinkable sleevings described in this specification sheet have not been flame retarded and will burn readily. These sleevings shall not be considered for acquisition when flame resistance is required. Additionally, only the outer jacket of Classes 2 and 3 sleevings are flame retardant.

Intended use: Heat shrinkable dual-wall polyolefin sleeving is used in one-step potting, encapsulating, or moisture sealing of electrical components.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS

(Project 5970-0577-04)

Review activities:
Army - MI, MU
Navy - EC
Air Force - 99
DLA - GS

User activities:
Army - ME
Navy - MC

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